

Industry Benchmark Review

Print-To-Frame ERP System

Source reviewed: github.com/madhukagamage6/Print-To-Frame-ERP-System (main branch, code-level review)

1. Executive Summary

Print-To-Frame is a React/Vite/Firebase single-page ERP application covering CRM (leads, deals, customers, partners), invoicing, a fabrication/work-order pipeline, logistics, an agent database, a cost calculator, messaging, and an admin panel with role-based permissions, built on a defined dark-theme design-token system ("Kinetic Blueprint").

Overall maturity: the core transactional workflow (lead → quote → job → invoice) is present and coherent, and the dark-first visual design already matches where 2026 power-tool UI trends are heading. Most modules are thinner than the industry-standard feature set for their category, though — particularly quotation governance, payment automation, permission granularity, and a few modern dashboard/navigation conventions.

Findings by priority across all modules and the UI/design review: Critical 3, High 6, Medium 11, Low 7.

2. Methodology

This review was based on a static, code-level read of the public GitHub repository (component structure, service layer, Firestore rules, design tokens, and permission logic). No authenticated live walkthrough of the deployed app was performed, so runtime interaction details (exact animations, modal transitions) are noted as inferred from code rather than observed directly.

Industry-standard comparisons were built from current (2026) product documentation and best-practice guides for each module category — CRM pipeline tools, invoicing platforms, RBAC guidance — plus current SaaS/admin-dashboard design-trend sources for the UI/design review. Full citations are listed in Section 7.

3. Stakeholder Impact Overview

The same gaps land differently depending on who's using the system. This section summarizes what matters most to each audience; Section 4's tables tag every item with who it affects.

Business / Owner

The pipeline and work-order tracking give good day-to-day visibility, but oversight tools are thin: there's no aging/revenue report, no permission-change audit trail, and quotations aren't structured enough to reliably report on. The three Critical items (structured quotations, payment collection, action-level permissions) are the ones most likely to cost money or create risk if left unaddressed.

Employee / Support / Ops

Staff get a real Kanban-driven workflow and a working permissions model, which is more than many small-business tools offer. The friction points are day-to-day: no bulk actions on leads, no list view for quick sorting, no stale-lead reminders, and re-quoting a customer currently means redoing the AI draft rather than revising a version.

Client / Customer

Clients experience the business mainly through quotes, invoices, and (per the work-order code) WhatsApp status updates. The most visible gap for them is having to pay via manual bank-transfer confirmation rather than an online payment link, and having no formal accept/reject step on quotations.

Agent / Partner

A Partners module exists in the CRM (Partners.jsx), but the review did not find a distinct partner-facing portal or view — partners appear to be managed as internal records rather than given their own access surface. Confirm whether partners need self-service visibility (e.g., into their own referred leads or commissions); if so, that's a design question for the business rather than a fixable gap.

4. Module-by-Module Findings

4.1 CRM — Leads / Deals Pipeline (Kanban)

Current state: Leads.jsx implements a genuine drag-and-drop-capable Kanban board (KanbanColumn/KanbanCard components) with stage-based columns (Intake through Completed), stage-color coding, and forward/back stage transitions. Leads can be converted into fabrication jobs directly from the pipeline, and stage changes are written back to Firestore.

Industry comparison: modern pipeline tools (Pipedrive, HubSpot, Zoho CRM) converge on a card format that surfaces deal value, assigned owner, and days-in-stage directly on the card, paired with a sortable/filterable list view for bulk work and configurable multi-pipeline support.

Item	Status	Priority	Affects	Why it matters
Card surfaces owner, value, days-in-stage at a glance	Partial	High	Owner, Ops	Cards currently emphasize status/customer info; without days-in-stage and assigned-owner visible on the card face, managers must open each card to spot stalled leads.
Sortable/filterable list view alongside Kanban	Missing	Medium	Ops	Kanban is good for flow visibility but poor for bulk edits, exports, and precise sorting — every mainstream pipeline tool pairs it with a list view.
Bulk actions (reassign, bulk stage change, export)	Missing	Medium	Ops	As lead volume grows, one-at-a-time updates become a bottleneck for sales/ops staff.
Configurable multi-pipeline / custom stages	Missing	Low	Owner, Ops	Single hard-coded stage list is fine at current scale but limits future product-line or workflow variation without a code change.
Stale-lead / follow-up reminders	Missing	Medium	Owner, Ops	Without automatic flags for leads stuck too long in a stage, follow-ups depend entirely on staff memory.

4.2 Client / Customer Database

Current state: Customers.jsx provides a searchable, filterable client registry, categorizes customers (Business/Individual), and cross-references related leads, invoices, and projects per customer.

Industry comparison: standard customer-database tooling additionally supports data export, a durable per-customer activity timeline, and duplicate detection.

Item	Status	Priority	Affects	Why it matters
CSV/data export of customer list	Missing	Medium	Owner, Ops	Common need for offline reporting, mail-merge, or migration; absence forces manual copy-paste.
Unified interaction timeline per customer	Partial	Medium	Ops	Related leads/invoices/projects are shown, but there's no single chronological activity feed — staff must check three tabs to reconstruct history.
Duplicate detection / merge tooling	Missing	Low	Ops	Not urgent at current scale, but duplicate customer records become a data-quality problem as the client base grows.

4.3 Quotation Generation

Current state: Quotation drafting is embedded inside the Lead detail view (LeadCardDetails.jsx), generated via an AI call seeded from job scope and pricing metadata, then stored as a draft string on the lead record.

Industry comparison: standalone quotation tooling in comparable systems is a first-class object with structured line items, versioning, and direct convert-to-invoice, sometimes with client-facing acceptance.

Item	Status	Priority	Affects	Why it matters
Structured line-item quote record (not free-text draft)	Missing	Critical	Owner, Ops, Client	A quotation stored as a generated text blob can't be reliably reused for invoicing, reported on, or audited — the single biggest structural gap found in the review.
Quote versioning history	Missing	High	Owner, Ops	Without versioning, re-quoting a customer overwrites the prior figure, losing the audit trail of what was actually offered and when.
Direct quote → invoice conversion	Missing	High	Owner, Ops	Invoices appear to be generated separately rather than derived from the accepted quote, risking figures drifting between the two.
Client-facing accept/reject (e-signature or link)	Missing	Low	Client	Reduces back-and-forth, though not core to the current manual sales process.

4.4 Invoice Generation & Synced Storage

Current state: Invoices.jsx supports Paid/Unpaid status filtering, PDF-style print output, and marking invoices Paid, with advance (75%) and final-settlement invoice types tailored to the framing business's own payment structure — a deliberate design choice, out of scope to change.

Industry comparison: standard invoicing platforms additionally provide integrated online payment collection, automated overdue reminders, and aging/revenue reporting, while keeping whatever payment-schedule structure the business already uses.

Item	Status	Priority	Affects	Why it matters
Online payment gateway integration (card/bank transfer)	Missing	Critical	Client, Owner	Payments currently rely on manual confirmation slips emailed to billing; this is the largest source of collection delay and manual reconciliation work relative to industry norms.
Automated overdue-payment reminders	Missing	High	Owner, Ops, Client	Every mainstream invoicing tool automates this because manual follow-up is the most commonly skipped operational task.
Aging report / revenue-by-period reporting	Missing	Medium	Owner	Without this, assessing cash-flow health requires manually scanning the invoice list.
Consistent invoice audit trail (create/edit/paid)	Partial	High	Owner	A generic audit-log service exists but it isn't evident every invoice status change is consistently logged through it — worth confirming and enforcing.

4.5 Work Order Management (Fabrication Pipeline)

Current state: FabricationWorks.jsx runs a genuine status pipeline (Pending → Ongoing → Ready For Inspection, etc.) with staff assignment, and an AI-drafted WhatsApp status-update message tied to job stage — a distinctive, business-appropriate touch worth keeping.

Industry comparison: work-order tools typically add attachment support for specs/photos, time/cost tracking, and automatic linkage forward into invoicing once a job completes.

Item	Status	Priority	Affects	Why it matters
Attachments (design files, progress photos) on a job	Missing	Medium	Ops, Client	For a custom-framing business, visual proof of work at each stage is a common trust-and-QA feature in comparable field-service/fabrication tools.
Time/cost tracking against the work order	Missing	Low	Owner	Useful for margin analysis per job, but not blocking day-to-day operations at current scale.
Auto-linkage from completed job to invoice creation	Partial	Medium	Owner, Ops	Invoicing appears to happen as a separate step rather than being triggered off job completion, risking missed or delayed billing.

4.6 User Access Management & Admin Panel

Current state: PermissionsManager.jsx implements a real per-role, per-module access matrix (nine roles), stored via PermissionsContext and enforced through a canAccess() check — a stronger baseline than many small business apps have.

Industry comparison: RBAC best practice for 2026 additionally expects action-level permissions, a visible audit log of permission changes, and a formal invite/deactivate lifecycle.

Item	Status	Priority	Affects	Why it matters
Action-level permissions (view/create/edit/delete/export)	Missing	Critical	Owner, Ops	Current model is per-module access level only; without action-level granularity, any role given module access likely has broader edit/delete power than intended — the most common RBAC gap flagged by security guidance.
Audit log of permission/role changes	Missing	High	Owner	Security best practice specifically calls out logging who changed a role's permissions and when — this is what auditors ask for first.
Formal user invite / deactivate flow	Missing	Medium	Owner, Ops	Without an invite/deactivate lifecycle, offboarding relies on someone remembering to edit or delete the right record.
Session management (view/force-logout)	Missing	Low	Owner	A more mature/enterprise feature; reasonable to defer at current team size.

5. UI / Visual Design Review

Current state: the app uses a defined design-token system ("Kinetic Blueprint": dark surface palette, Hanken Grotesk/Poppins typography scale, consistent primary/secondary/error tokens) applied via Tailwind, plus a single icon library (lucide-react) and purpose-built loading/empty-state components. This is a genuinely dark-first design — which happens to already match where 2026 SaaS/admin-tool design is heading.

Industry comparison: 2026 SaaS dashboard convention (Linear, Vercel, Supabase, Attio) favors modular, rearrangeable dashboard widgets with a single global filter, command-palette navigation, restrained use of color reserved for status meaning, and dark mode designed alongside a light-mode option rather than as the only theme.

Item	Status	Priority	Affects	Why it matters
Modular/rearrangeable dashboard widgets with a global filter	Missing	Medium	Owner, Ops	2026 SaaS dashboard convention lets each user tailor what they see and filter every widget at once; the current dashboard is fixed-layout.
Command palette (Cmd/Ctrl+K) for fast navigation	Missing	Low	Ops	Increasingly standard in power tools (Linear, Vercel, Supabase) for keyboard-first navigation across a growing number of modules.
Full responsive breakpoints on data-heavy screens (Kanban, tables)	Partial	Medium	Ops, Client	Some tailwind breakpoint usage exists but is thin on the densest screens (Kanban board, wide tables); these are the screens most likely to break on tablet/mobile.
Light-mode option alongside the existing dark theme	Missing	Low	Ops, Client	The app already does dark-mode-first well, which matches 2026 power-tool convention — but competitors increasingly ship both themes as user preference, not just one.

6. Prioritized Action Plan

Ordered Critical → Low across all modules and the UI/design review. Each plan extends the existing stack (React + Tailwind + Firebase/Firestore) rather than introducing new infrastructure.

Critical

1. Convert quotations from AI-generated text drafts into structured records.

- Add a `quotations` Firestore collection with line items (description, qty, unit price, tax, discount), a status field (Draft/Sent/Accepted/Rejected), and a version number.
- Keep the existing AI drafting step, but have it populate these structured fields instead of a single text blob.
- Add a 'New Version' action that clones the current quote into a new version rather than overwriting it.
- Add a 'Convert to Invoice' action that copies accepted line items directly into a new invoice record.

2. Add online payment collection to invoicing.

- Integrate a payment gateway appropriate to the business's market behind the existing 'Print PDF Invoice' action.
- Add a client-facing payment link included on the emailed/PDF invoice.
- On successful payment webhook, auto-update invoice status to Paid and log the event via the existing auditLog.js service.

3. Add action-level permissions to the access-management model.

- Extend the permissions schema from `{ role: { module: accessLevel } }` to `{ role: { module: { view, create, edit, delete, export } } }`.
- Update `canAccess()` and every call site to check the specific action being performed, not just module visibility.
- Update PermissionsManager.jsx's matrix UI to expose per-action toggles instead of a single access-level dropdown per cell.

High

4. Log every invoice and permission change explicitly through auditLog.js, and add automated overdue-payment reminders via the existing messaging integration.

5. Add quote versioning history display and a direct quote→invoice conversion action, reusing the schema from Critical #1.

6. Surface owner/value/days-in-stage on Kanban cards, and add a permission-change audit log view in the admin panel.

Medium

7. Add a sortable/filterable list view alongside the Kanban board, reusing the existing filter bar component.

8. Add stale-lead reminders, customer CSV export, a unified customer activity timeline, an invoice aging report, attachment support on work orders, auto-linkage from completed work orders to invoicing, and modular/rearrangeable dashboard widgets with a global filter.

Low

9. Configurable multi-pipeline support, customer duplicate-detection, work-order time/cost tracking, admin session management, client-facing quote e-signature, a command palette, and a light-mode theme option alongside the current dark theme.

7. Sources

- [Onesuite — 7 CRMs with Pipeline Management That Actually Close Deals \(2026\) — onesuite.io/blog/crm-and-pipeline-management](#)
- [OnePageCRM — In-Depth Pipeline CRM Review 2026 — onepagecrm.com/crm-reviews/pipeline](#)
- [CRM.org — What Is Kanban? How CRMs Use It — crm.org/news/what-is-kanban-methodology](#)
- [Alphonso Labs — 15 Essential CRM Features to Check Before You Buy — alphonsolabs.com/crm-essential-features-checklist-2026](#)
- [The Digital Project Manager — 10 Best Invoicing Software in 2026 — thedigitalprojectmanager.com/tools/best-invoicing-software](#)
- [Salesforce — 10 Best Invoicing and Billing Software for 2026 — salesforce.com/sales/revenue-lifecycle-management/billing-software](#)
- [Aviy — Features to Look for in Invoice Software \(2026 Buyer's Checklist\) — aviy.ai/blog/features-to-look-for-in-invoice-software](#)
- [CloudEagle — Role-Based Access Control: Implementation, Examples & Best Practices — cloudeagle.ai/blogs/role-based-access-control](#)
- [Aerospike — RBAC Guide for Modern Data Security — aerospike.com/blog/role-based-access-control-rbac-guide](#)
- [Pathlock — Role-Based Access Control \(RBAC\): A Comprehensive Guide — pathlock.com/blog/role-based-access-control-rbac](#)
- [SaaSFrame — The Anatomy of High-Performance SaaS Dashboard Design: 2026 Trends & Patterns — saasframe.io/blog/the-anatomy-of-high-performance-saas-dashboard-design-2026-trends-patterns](#)
- [AdminLTE.IO — 10 Best SaaS Dashboard Design Examples & Trends \(2026\) — adminlte.io/blog/saas-dashboard-design-examples](#)
- [925 Studios — 35 SaaS Dashboard Design Examples, Trends and Patterns \(2026\) — 925studios.co/blog/saas-dashboard-design-examples-2026](#)

8. Quick-Reference Checklist

Every finding from this report, grouped by priority. Check a box to mark that the business has decided to implement that item; leave it unchecked (or mark with a cross) to record a decision not to.

Critical

- ✓ [Quotation Generation] Structured line-item quote record (not free-text draft)
- ✓ [Invoicing] Online payment gateway integration (card/bank transfer)
- ✓ [Access Management] Action-level permissions (view/create/edit/delete/export)

High

- ✓ [CRM — Leads/Deals] Card surfaces owner, value, days-in-stage at a glance
- ✓ [Quotation Generation] Quote versioning history
- ✓ [Quotation Generation] Direct quote → invoice conversion
- ✓ [Invoicing] Automated overdue-payment reminders
- ✓ [Invoicing] Consistent invoice audit trail (create/edit/paid)
- ✓ [Access Management] Audit log of permission/role changes

Medium

- ✓ [CRM — Leads/Deals] Sortable/filterable list view alongside Kanban
- ✓ [CRM — Leads/Deals] Bulk actions (reassign, bulk stage change, export)
- ✓ [CRM — Leads/Deals] Stale-lead / follow-up reminders
- ✓ [Customer Database] CSV/data export of customer list
- ✓ [Customer Database] Unified interaction timeline per customer
- ✓ [Invoicing] Aging report / revenue-by-period reporting
- ✓ [Work Orders] Attachments (design files, progress photos) on a job
- ✓ [Work Orders] Auto-linkage from completed job to invoice creation
- ✓ [Access Management] Formal user invite / deactivate flow
- ✓ [UI/Design] Modular/rearrangeable dashboard widgets with a global filter
- ✓ [UI/Design] Full responsive breakpoints on data-heavy screens (Kanban, tables)

Low

- X [CRM — Leads/Deals] Configurable multi-pipeline / custom stages
- ✓ [Customer Database] Duplicate detection / merge tooling
- x [Quotation Generation] Client-facing accept/reject (e-signature or link)
- x [Work Orders] Time/cost tracking against the work order
- x [Access Management] Session management (view/force-logout)
- x [UI/Design] Command palette (Cmd/Ctrl+K) for fast navigation
- ✓ [UI/Design] Light-mode option alongside the existing dark theme